

CMSC 104 Section 2

Fall 2026

Sample Quiz 2

Instructions:

This quiz is to be completed on paper, using a pen or pencil. Please make sure you write your name on the paper. When you are finished you may turn in your paper and leave the room.

This quiz is “open-book, open-notes.” You MAY use any of the following: your laptop or phone, a book; your notes; the instructor’s notes and code from the GitHub site; an IDE such as IDLE or PyCharm (if you wish to test your code to make sure your answer is correct). You MAY NOT use: any other person, whether in this classroom or not (e.g., you may not text your friend the computer science major), or any AI-support source such as ChatGPT, Gemini, CoPilot, Claude Code, etc. If you have a question about this, please raise your hand and ask.

(Note to me: I need to add three “while” loop questions to this.)

Section 1: True/False and Multiple Choice (40 points)

There are 10 questions in this section. Each is worth 4 points. No partial credit is given for any question in this section. If you change your answer during the quiz, please make sure you clearly indicate which is your final answer.

1. How are function arguments (in the function call) and function parameters (in the function definition) matched up?
 - a. Anything with the same name is matched up. If the second argument is “x” it will be matched with the fourth parameter also named “x.”
 - b. In order. It doesn’t matter what variable names you use; the first argument gets matched with the first parameter; the second with the second; etc.
 - c. In some random order that only Guido van Rossum, the inventor of Python, understand
 - d. None of the above is correct

2. Which of the following is a valid Python variable name?
 - a. 4dyas2SpringBreak
 - b. Fore_Days_2_Spring_break
 - c. Four-Days-2-Spring_break
 - d. None of these is a valid Python variable name

3. True or False: in Python, an argument used in a function call can be a literal. That is, you can call a function with:

```
y = calculate_volume(radius, height, 3.14159)
```

- a. True
 - b. False
4. True or False: a “for i” loop is the most general loop in Python. That is, any problem that can be solved with a loop can be solved with a “for i” loop, but that same statement is not true for “while” loops.
- a. True
 - b. False
5. Is Microsoft Excel hardware or application software?
- a. Hardware
 - b. Application software

6. What does the following Python code produce as output?

```
product = 1
for i in range(1,6,1):
    product = product * i

print(product)
```

- a. 120
 - b. 720
 - c. 6
 - d. None of the above answers is correct
7. In general, what is the best way to develop computer software?
- a. Write a little; test a little
 - b. Write the whole program at once and hope like heck it works
 - c. Just steal it from somebody else and hope they don't sue you for theft
 - d. Seriously? Just choose a.
8. True or False: you can have an if-elif-else statement without an else. That is, the statement

```
if age > 21:
    print("You have reached the age of majority")
elif age > 18:
    print("You can vote but you are not fully an adult")
```

is legal in Python

- a. True
- b. False

9. What is the value printed by the following line of code?

```
print ((8%4)> 0 and 5//2 < 3**2)
```

- a. True
- b. False
- c. 1
- d. None of the above is correct

10. Which of the following conditions have to be true in order for you to import a Python program you wrote into another Python program you are writing?

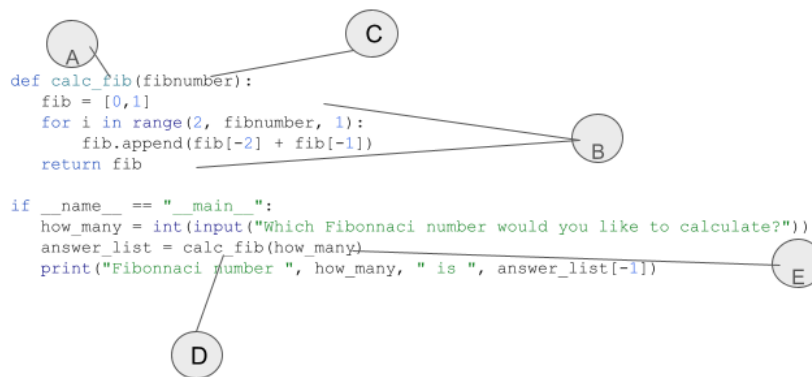
- a. You have to know where the program is - that is, Python has to somehow be told what directory the imported program can be found
- b. The program has to have a name that ends in .py
- c. The program can't contain more than 4 functions
- d. A and B are both correct

Section 2: Short answer (40 points)

There are 8 questions in this section. Each is worth 5 points. Partial credit IS given for questions in this section, so please make an attempt to answer each question.

Note: “short answer” means one or two sentences. It should not take you more than that to answer a question. Continuing to write paragraph after paragraph is probably not going to help make your answer more correct.

11. Identify the parts of a function definition and call



Argument(s)
Parameter(s)
Function call
Function body
Function name

12. Explain what a “sentinel while loop” is in Python.

13. Write a “for i” loop that computes the product of the first 10 integers- the numbers from 1 to 10. .

14. What is a Boolean variable?

15. What is the output produced by running the following Python code?

```
count = 5
while count > 0:
    print(count, end=" ")
    count -= 2
```

16. The syntax of the statement that starts a Python “for i” loop is

```
for i in range (start, stop, step):
```

What does “step” mean? What does it cause to happen?

17. Write an if-elif-else statement that does the following:

- Suppose that x is an integer variable and y is a string variable
- If $x > 0$ and y is not equal to “error” print “the run was successful”
- If $x = 0$ and y is equal to “error” print “the results are inconsistent”
- In all other cases, print “the program encountered an error; please try again later”

18. Write two lines of code that first import the Python math library, and then use the sin() function to print out the sine of 2π radians

Section 3: Programming (20 points)

There are two questions in this section; each is worth 10 points. Partial credit IS given for questions in this section.

You do not need to write comments with your code, unless you think that it will help the grader more clearly understand exactly what you're trying to accomplish.

19. Write a Python program that will generate a series of random numbers and add them up. You must first import the random library. Then, ask the user how many random numbers they want to add. Set the sum of the numbers to zero. Then write a for I loop that runs to the number the user entered.

20. Write a Python function that calculates the volume of a right circular cone. The formula is $v = \pi r^2 h / 3$. Your function should have two parameters, the radius and the height. It should have one return, the volume